

DevManchego Chess

User Manual

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1. Introduction

DevManchego Chess is a chess application with a graphical interface developed in Java, offering both games between two players and games against artificial intelligence. The application includes an advanced analysis panel, training modes, and special battle scenarios with non-standard armies.

1.1 Main Features

- Single-player games (vs. AI) and 2-player games on the same computer.
- Five difficulty levels: Beginner, Easy, Medium, Hard, and Expert.
- Choice of piece color (White or Black) with automatic board flip.
- Analysis panel with Principal Variation, move history, and move quality ratings.
- Real-time captured pieces panel.
- Training Mode: Checkmate Drills and Special Battles.
- Undo moves, save and load games, export as PGN and image.
- Interface available in Spanish, English, French, German, and Italian.

System Requirements

- Operating System: Windows 10/11, Linux, or macOS
- Java Runtime Environment (JRE) 17 or higher
- Minimum recommended resolution: 1280 × 800 pixels

2. Application Startup

When you launch DevManchego Chess, the Welcome Dialog appears automatically, where you configure the basic game options before starting.

2.1 Welcome Dialog

The welcome window presents the following options:

Game Mode

- 1 Player — Game against artificial intelligence.
- 2 Players — Game between two people on the same computer.

Difficulty Level (1-player mode only)

Select the level from the dropdown menu:

Level	Description
Beginner	AI plays unpredictably (40% random moves). Ideal for learning.
Easy	Play with occasional errors (20% randomness). Recommended for beginners.
Medium	Balanced AI with no gross errors. Suitable for intermediate players.
Hard	Solid AI with no randomness. Default level. Presents a real challenge.
Expert	Maximum search depth. For advanced players.

Piece Color Choice

In 1-player mode you can choose whether you want to play with White or Black pieces. If you choose Black, the board automatically rotates so your pieces appear at the bottom.

Do Not Show Again

If you check 'Do Not Show Again,' the dialog will not appear when starting the application in future sessions. You can always return to it through File → New Game.

☐ Note: If you close the welcome dialog without confirming, the application starts with default settings: 1 player, Hard level, White pieces.

3. Main Interface

The main window of DevManchego Chess is divided into two areas:

Area	Description
Chess Board (left)	Interactive play area. Occupies approximately 2/3 of the total window width.
Analysis Panel (right)	Displays game information: engine status, principal variation, captured pieces, and move history.

3.1 Chess Board

The board follows standard algebraic notation. Coordinates (letters a–h and numbers 1–8) are displayed on the edges. If you play with black pieces, the board appears rotated with black at the bottom.

How to Move Pieces

12. Click on the piece you want to move. The square will highlight in yellow.
13. Available legal moves are shown with purple circles.
14. Click on the destination square to execute the move.

If you click on another piece of yours, the selection changes. If you click outside the board or on an invalid square, the selection is cancelled.

3.2 Status Bar

Just below the board is a status label indicating whose turn it is, if there is check, checkmate, draw, or if the AI is calculating its move.

3.3 Analysis Panel

The right panel contains the following sections:

Move Quality History

The top shows the accuracy percentage for each player and a colored dot for each move made:

Color	Meaning	Rating
☒ Bright Green	Best Move	Loss < 5 cp
☒ Teal Green	Excellent	Loss 6–20 cp
☐ Blue	Good	Loss 21–50 cp
☐ Yellow	Inaccuracy	Loss 51–100 cp
☐ Orange	Mistake	Loss 101–200 cp
☐ Red	Blunder	Loss > 200 cp

Engine Information

Shows the engine in use (Engine: Java AI), the search depth, and the position score in centipawns (positive = white advantage, negative = black advantage).

Principal Variation (PV)

When you press the Analyze button, the sequence of moves that the engine considers optimal is displayed. Each line shows the move number and UCI notation for each move.

Captured Pieces

Shows the Unicode symbols of pieces each side has captured during the game, ordered by value (Queen, Rook, Bishop, Knight, Pawn). Updates with each move.

Move History

Complete record of all moves in the game in UCI notation (e.g.: e2e4 e7e5). Automatically expands with each move.

4. Menus

4.1 File Menu

Option	Description
New Game	Opens the configuration dialog to start a new game. If there is a game in progress, confirmation is requested.
Save Game	Saves the current game state to a .chess file for later resuming.
Load Game	Loads a previously saved game.
Export PGN	Exports the move history in standard PGN format, compatible with other chess programs.
Export Image	Saves a screenshot of the current board as a PNG image.
Exit	Closes the application.

4.2 Game Menu

Option	Description
Undo Move	Undoes the last move (or last two in 1-player mode to return to the player's turn).
Difficulty	Submenu with five difficulty levels. Change applies immediately to the AI's next move.
MultiPV	Adjusts how many alternative variations the engine calculates when analyzing (1–4).

4.3 Training Menu

The Training menu contains two subcategories: Checkmate Drills and Special Battles. Both are described in detail in sections 6 and 7.

4.4 Language Menu

Allows you to change the interface language in real-time without restarting. Available languages: Spanish, English, French, German, and Italian.

5. Analysis Panel

The analysis panel provides real-time information about move quality and engine status. Its use is optional and does not interfere with the flow of the game.

5.1 Analyze Button

Starts a deep analysis of the current position. The engine evaluates the position through iterative search and displays the Principal Variation. During analysis, the button changes to Stop to allow interruption.

5.2 Auto Mode

When the Auto checkbox is checked, analysis is launched automatically after each move without needing to press Analyze manually.

5.3 MultiPV

The MultiPV selector (1–4) indicates how many different game lines the engine calculates in each analysis. A higher value provides more alternatives but reduces the depth reached in the same time.

5.4 Best Move Arrow

When the engine has completed the analysis, a green arrow is drawn on the board indicating the move it considers optimal. This arrow disappears when you make the next move.

☐ Tip: Enable Auto mode and play normally. After each move, the green arrow will show you if there was a better option. Excellent way to learn without interrupting play.

6. Checkmate Drills

Checkmate Drills are practice positions where White must checkmate the Black king (which has only its king) with specific material. Ideal for learning fundamental endgame patterns.

Access: Training Menu → Checkmate Drills → select the type.

Drill	Description
King and Queen vs. King (KQ vs K)	Checkmate achieved in few moves. Ideal for getting familiar with basic endgames.
King and Rook vs. King (KR vs K)	Slightly more complex than the queen. Requires more careful technique to trap the enemy king.
King and Two Bishops vs. King (KBB vs K)	Two bishops must work together to push the king to the board edge.
King, Bishop and Knight vs. King (KBN vs K)	Most difficult of the basic endgames. Requires driving the enemy king to the corner of the bishop's color.

How to Use Drills

15. Select the desired drill. The board configures automatically.
16. White must checkmate the lone black king.
17. The AI plays as the black king and tries to escape.
18. Use the Analyze button to get help from the engine if stuck.
19. To restart, select the drill again from the menu.

7. Special Battles

Special Battles are games with non-standard armies: each side receives an atypical set of pieces that creates unique game dynamics. Fun for experimenting with asymmetrical material advantages.

Access: Training Menu → Special Battles → select battle type.

7.1 Battle Setup Dialog

When you select a battle, a dialog appears where you must choose:

- Side A or Side B — each side has a different army. Blue and red cards show the army name.
- Piece color — whether you want to play with White or Black pieces.
- Click Start to begin the battle.

7.2 Battle Types

Knight Duel

Side A (Knights)

- 10 knights: 2 flanking the king + 8 on the second rank.

Side B (Bishops + Pawns)

- 6 bishops on the back rank + 12 pawns in staggered formation.

Heavy Artillery

Side A (Rooks)

- 5 rooks: 2 on the first rank corners + 3 on the second rank.

Side B (Queens + Pawns)

- 3 queens flanking the king + 3 shield pawns.

Pawn Wall

Side A (Pawns)

- 24 pawns distributed in three complete ranks.

Side B (Knights + Bishops)

- 4 knights and 4 bishops in symmetric formation.

Minor Piece Clash

Side A (Knights)

- 8 knights: 2 flanking the king + 6 on the second rank.

Side B (Bishops)

- 8 bishops: 2 flanking the king + 6 on the second rank.

☐ Warning: Standard chess rules apply in special battles (check, checkmate, draw).

The only difference is the non-standard initial material for each side.

8. Shortcuts and Quick Gestures

The application can be fully controlled with the mouse. Below is a summary of the most frequent actions:

Action	Effect
Click on a piece	Selects the piece and shows legal moves.
Click on highlighted square	Executes the move.
Click on another own piece	Changes selection to that piece.
Click outside the board	Cancels current selection.
Analyze button	Starts engine analysis of current position.
Stop button	Interrupts current analysis.
Auto checkbox	Toggles automatic analysis after each move.
Game Menu → Undo	Undoes the last pair of moves (player + AI).

9. Saving and Loading Games

9.1 Save Game

20. Open the File menu.
21. Select Save Game.
22. Choose the location and filename in the dialog.
23. Click Save. The file is stored with the .chess extension.

9.2 Load Game

24. Open the File menu.
25. Select Load Game.
26. Navigate to the previously saved .chess file.
27. Click Open. The game is restored to its exact saved state.

9.3 Export as PGN

PGN format (Portable Game Notation) is compatible with most chess programs (Lichess, Chess.com, Arena, etc.). To export:

28. File Menu → Export PGN.
29. Select where to save the .pgn file.
30. Click Save.

9.4 Export Board Image

Save a PNG image of the current board, useful for sharing positions.

31. File Menu → Export Image.
32. Select location and filename.
33. Click Save.

10. Frequently Asked Questions

Can I change difficulty during a game?

Yes. Go to Game Menu → Difficulty and select the desired level. The change applies to the AI's next move.

Why doesn't a promotion button appear when a pawn reaches the last rank?

The application automatically promotes the pawn to a queen when it reaches the last rank.

What happens if I close the new game dialog without confirming?

If you close the dialog by clicking the X or pressing Escape, the operation is cancelled and the current game is not modified.

Can I play with Black pieces?

Yes. In the new game dialog (or Special Battles dialog), select the Black button. The board will rotate so your pieces are at the bottom.

Why does the move quality panel take time to update?

Classifying each move requires a quick background analysis. The colored dots appear within tenths of a second after each move.

Does the confirmation dialog appear even after checkmate?

No. The confirmation window only appears if the game is in progress. If the game has ended (checkmate or draw), the new game starts without asking.

How do I restore the welcome dialog if I disabled it?

Use File Menu → New Game. This always shows the configuration dialog regardless of your saved preference.

11. Glossary

Term	Definition
Centipawn (cp)	Unit of measure of positional advantage. 100 cp roughly equals the value of one pawn.
UCI	Universal Chess Interface. Standard protocol for communication between chess programs.
PGN	Portable Game Notation. Text format for recording chess games.
Principal Variation (PV)	Sequence of moves the engine considers optimal for both sides.
MultiPV	Number of alternative variations the engine calculates simultaneously.
Depth	Number of half-moves (plies) the engine analyzes.
En passant	Special pawn capture in chess. Automatically applied.
Castling	Special move between king and rook. Execute by moving the king two squares toward the rook.
Blunder	Move causing a loss of more than 200 centipawns.
Drill	Practice exercise with a specific position to train a particular pattern.

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Enjoy the game!